

```
In [ ]: if
        if-else
        if-elif-else
```

```
In [ ]: # Two classifications
        # you want to checkout some thing based on some condition

        # in python in syntax wherever we have : there is an indentation
        # indentation is space
        # your code of lines will start from that point
```

```
In [ ]: if <condition>:
        ----- if area
        ----- if area
        ----- if area

        ----- out of if area
```

```
In [ ]: name=input("enter a name:")
        if name=='python'
```

```
In [ ]: ##### assign #####
        name='python' # we are saving a value 'python' in a variable 'name'

        ##### Condition#####
        name=='python' # we are checking the name is equal to python or not
```

```
In [1]: 'hello'=='python'
```

Out[1]: False

```
In [8]: "python"=="python"
```

Out[8]: True

```
In [7]: # M-1
        if "python"=="python":
            print("hai")
            print(1)
            print("good")
```

```
In [9]: # M-2
        if True:
            print("hai")
            print(1)
            print("good")
```

```
hai
1
good
```

```
In [12]: name=input("enter a name:") # we are saving a name in a variable
if name=="python":                # checkin the condition True and False
    print("hai")
    print(1)
    print("good")
```

enter a name:hello

```
In [ ]: if "python"=="python":
if True
if name=="python":
```

```
In [16]: name=input("enter a name:") # we are saving a name in a variable

#####

if name=="python":                # checkin the condition True and False
    print("hai")
    print(1)
    print("good")

#####
else:
    print("enter a valid name")
```

enter a name:hello

enter a valid name

```
In [17]: print('hey')
name=input("enter a name:")
print('good morning')
#####
if name=="python":
    print("hai")
    print(1)
    print("good")

else:
    print("enter a valid name")
    print('sorry')

#####3

print("now its fine")
```

hey

enter a name:python

good morning

hai

1

good

now its fine

```
In [21]: print('hey')
name=input("enter a name:")
print('good morning')
#####
if name=="python":
    print("hai")
    print(1)
    print("good")

#print("now else statement starts")

else:
    print("enter a valid name")
    print('sorry')

#####3

print("now its fine")
```

```
hey
enter a name:hello
good morning
now else statement starts
now its fine
```

```
In [22]: 9/5      # divison Normal divison value
```

```
Out[22]: 1.8
```

```
In [23]: 9//5     # floor division que
```

```
Out[23]: 1
```

```
In [24]: 9%5      # modulus      reminder
```

```
Out[24]: 4
```

```
In [25]: # WAP ask the user enter a number
# print if that number is even or odd

# even: when you divide any number with 2 , the remiander shoudl be zero
# step-1: num=eval(input("enter a number"))
# step-2: if <condition>:
# step-3     print("it is an even")
# step-4: else:
# step-5:     pront("it is an odd number")

num=eval(input("enter a number"))
if num%2==0:
    print("it is an even")
else:
    print("it is an odd number")
```

```
enter a number10
it is an even
```

```
In [29]: num=eval(input("enter a number"))
if num%2==0:
    print(f"{num} is an even")
else:
    print(f"{num} is an odd number")
```

enter a number0
0 is an even

```
In [30]: 2/0
```

ZeroDivisionError

Traceback (most recent call last)

Cell In[30], line 1

----> 1 2/0

ZeroDivisionError: division by zero

- Name error (have you provided the variable above the lines)
- syntax error (.,}, {}, ,)
- indentation error (space after :)
- zero division error (something divided by zero)
- value error (related to math error)
- attribute error (method not identified from module/package)

```
In [45]: # WAP ask the user enter a random number between 1 to 50
# print it is even or odd
```

```
# step-1: import <package_name>
# step-2: num=<p_name>.<method>(1,50)
# step-3: if <condition>:
#step-4:     print("even")
# step-5: else:
# step-6:     print("odd")
```

```
import random
num=random.randint(1,50)
if num%2==0:
    print(f"{num} is even")
else:
    print(f"{num} is odd")
```

47 is odd

```
In [46]: # how you are passing the input
num=10
if num%2==0:
    print(f"{num} is even")
else:
    print(f"{num} is odd")
#####
num=eval(input())
if num%2==0:
    print(f"{num} is even")
else:
    print(f"{num} is odd")
#####

import random
num=random.randint(1,50)
if num%2==0:
    print(f"{num} is even")
else:
    print(f"{num} is odd")
```

```
10 is even
13
13 is odd
28 is even
```

```
In [48]: # wap ask the user enter a name
#      ask the user enter a salary
#      ask the user enter a tax percentage
#print he earns good if tax pay more than 1lakh
#else print he earns less

# step-1: name=input("enter a name:")
# step-2: salary=eval(input("enter salary:"))
# step-3: tax_per=eval(input("enter the tax in per:"))
# step-4: tax_pay=<>*<>/100
# step-5: if <cond>:
# step-6:   print(f"{name} eearns good")
# step-7: else:
# step-8:   print(f"{name} eearns Less")

name = input("enter a name: ")
salary = eval(input("enter a salary: "))
tax_percentage = eval(input("enter a tax in %: "))
tax_pay = (salary * tax_percentage)/100

if tax_pay >= 100000:
    print(f"{name} earns good")
else:
    print(f"{name} earns less")
```

```
enter a name: sourav
enter a salary: 50000000
enter a tax in %: 10
sourav earns good
```

```
In [49]: # wap ask the user get a random number between 1 to 200
# print value is greater than 99 , if the value greater than 99
# print value is less than 99, if the value less than 99

# step-1: import <pname>
# step-2: num=<>.<>(1,200)
# step-3: if num>99:
# step-4:     print("value is greater than 99")
# step-5: else:
# step-6:     print("value is less than 99")

num=random.randint(1,200);
if num>99:
    print(f"{num} is greater then 99")
else:
    print(f"{num} is less then 99")
```

9 is less than 99

```
In [52]: # wap ask the user enter a number (keyboard)
# if that number >=0 print (pos num)
# else: print(it is a negative number)
number=eval(input("Enter a number: "))
condition= number>0
if condition:
    print("Its a positive number")
else:
    print("Its a negative number")

>0
<0
=0
```

Enter a number: 0
Its a negative number

if-elif-else

```
In [4]: num=eval(input("enter the number:"))
#num>0 ==== > it is a pos
#num<0 ==== > it is a neg
#num==0 ===== > it is a zero

if num>0:
    print(f"{num} is a postive number")
elif num<0:
    print(f"{num} is a negative number")
else:
    print("it is zero")
```

enter the number:100
100 is a postive number

```
In [6]: #WAP ask the user enter a postive number
# if number =1 then print("one")
# if number=2 then print("two")
# if number=3 then print("three")
# if number >3 then print ("not valid")

# if-elif-elif-else

num=eval(input("enter a postive number:"))
if num==1:
    print("one")

elif num==2:
    print("two")

elif num==3:
    print("three")

else:
    print("invalid")
```

```
enter a postive number:2
two
```

```
In [13]: #WAP ask the user enter percentage of marks
# if percentage>90    print("A")
# if percentage between 75 to 90    print("B")
# if percentage between 50 to 75    print("C")
# if percentage between 35 to 50    print("D")
# otherwise print("Fail")

# if-elif-elif-elif-else

num=eval(input("enter the % of marks")) # num=60
if num >= 90:
    print(f"{num} is A grade")
elif num >= 75:
    print("B grade")
elif num >= 50 :
    print("C grade")
elif num >= 35 :
    print("D grade")
else:
    print("Fail")

# you have given 95
```

```
enter the % of marks90
90 is A grade
```

```
In [14]: perc=eval(input("enter percentage"))
if perc>90: # 40>90    F
    print(f"A grade")

elif perc>75 and perc==90:  40>75 and  40==90    F and F    F
    print("B")

elif perc>50 and perc==75:  40>50    and 40==75    F    and F    F
    print("C")

elif perc>35 and perc==50:  40>35 and 40==50    T    and F    F
    print("D")
else:
    print("fail")
```

enter percentage40
fail

```
In [ ]: A   B   AandB   AorB
0     0     0         0
0     1     0         1
1     0     0         1
1     1     1         1
```

```
In [15]: #WAP ask the user enter an age
# if age>60 print("senior citizen")
# if age between 45 to 60    print("Middile aged")
# if age between 30 to 45    print("aged")
# if age between 20 to 30    print("young")
# if age between 13 to 20    print("teenage")
# if age Less than 13        print ("kid")

age=eval(input("enter the age:")) # age=35
if age>=60:print("Senior Citizen")
elif age>=45:print("MA")
elif age>=30:print("aged")
elif age>=20:print("youth")
elif age>=13:print("Teenage")
else:print("kid")
```

enter the age:30
aged


```
In [21]: # WAP ask the user enter a distance
# if the distance<5Km
# ask the user enter a fare: 25
# calculate the charge

# if the distance between 5 to 20k
# ask the user enter a fare:100
# calculate the total charge

# if the distance greater than 20km
# ask the user enter a fare:200
# calculate the tota charge

import time
distance=eval(input("enter the distance in km:")) # 3
if distance<5:
    print("user: how much fare")
    time.sleep(2)
    print("conductor:the distance is less than 5km so fare is 25rs")
    fare=eval(input("enter the fare:")) # 25
    charge=distance*fare
    print(f"the charge for {distance} km is {charge} rs") #

elif distance<20:
    print("the distance is less than 20km")
    fare=eval(input("enter the fare:")) # 100
    charge=distance*fare
    print(f"the charge for {distance} km is {charge} rs") #

else:
    print("the distance greater than 20km")
    fare=eval(input("enter the fare:")) # 200
    charge=distance*fare
    print(f"the charge for {distance} km is {charge} rs") #
```

```
enter the distance in km:3
user: how much fare
conductor:the distance is less than 5km so fare is 25rs
enter the fare:25
the charge for 3 km is 75 rs
```

```
In [22]: # ask the user enter a gender
# if gender == 'male'
#     ask the user enter your age
#     if age > 20: print ('man')
#     otherwise: print ("boy")

# elif gender=="female"
#     ask the user enter age
#     if age>20: print("woman")
#     otherwise: print("girl")

# else:
#     print('Fine')

gender=input("enter a gender:")
if gender=='male':
    age=eval(input("enter age:"))
    if age>=20:
        print("Man")
    else:
        print("boy")

elif gender=='female':
    age=eval(input("enter age:"))
    if age>=20:
        print("woman")
    else:
        print("girl")

else:
    print('Fine')
```

```
enter a gender:female
enter age:25
woman
```

```
In [ ]: #ask the user to genter gender. if gender = male then ask to enter the age

Gender = input("Please enter your Gender: ")
if Gender == "Male":
    Age = eval(input("Plese Enter your Age: "))
    if Age > 20:
        print("you are a Man")
    else:
        print("You are a Boy")

elif Gender == "Female":
    Age = eval(input("Plese Enter your Age: "))
    if Age > 20:
        print("you are a Woman")
    else:
        print("You are a Girl")
else:
    print("Fine")
```

```
In [ ]: # Step-1: hey mom
# Step-2: hello kid
# Step-3: Kid: Mom I got to know that Gove has implemented free bus service
# Step-4: Mom: Yes! I too hear that
# Step-5: Kid: Lets go to Grand mother house
# Step-6: Mom: Okay sure!
# Step-7: distance=Conductor : Enter a distance =====> Input
# Step-8: Conductor : show me Identity card
# Step-9: ans=Mom: NO
# Step-10 : If ans==NO
#           step-11: Conductor: You need to take the ticket
#           step-12: Mom: How much the fare
#           step-13: Conductor: you mentioned your distance 50km, for 50km t
#           step-14: fare=enter the fare
#           step-15: total charge
#step-16: If ans==Yes
#           Step-17: Conductor enjoy your free trip
```

```
In [25]: print("kid: hey mom")
print("mom: Hi,Kid")
print("kid: govt had implemented the free bus services for all females")
print("mom: I too heard about this news")
print("kid: lets goto grandmother house")
print("mom: yeah sure")
distance=eval(input("Conductor:enter the distance"))
print("Conductor: show me the identity card")
answer=input("mom: needto answer")
if answer == 'NO':
    print("Conductor: please take the ticket then")
    print("Mom: How much the fare?")
    print(f"Conductor: you mentioned the distance as {distance}kms. So the
charge=distance*100
else:
    print("You have issued with a free ticket")
```

```
kid: hey mom
mom: Hi,Kid
kid: govt had implemented the free bus services for all females
mom: I too heard about this news
kid: lets goto grandmother house
mom: yeah sure
Conductor:enter the distanceyes
```

```
-----
-
NameError                                Traceback (most recent call las
t)
```

```
Cell In[25], line 7
      5 print("kid: lets goto grandmother house")
      6 print("mom: yeah sure")
----> 7 distance=eval(input("Conductor:enter the distance"))
      8 print("Conductor: show me the identity card")
      9 answer=input("mom: needto answer")
```

```
File <string>:1
```

```
NameError: name 'yes' is not defined
```

```

In [26]: print("hey mom")
print("hello kid")
print("kid: Mom I got to know that govt. has implemented free bus services")
print("Mom: Yes! i too hear that")
print("kid: lets go to grand mothers house")
print("Mom: ok, sure")

distance = eval(input("I am conductor of this bus please let me know the di
idcard = input("You have the ID card 1. yes or 2. No Show me your Id card")

if idcard == 'No' or 'no':
    conductor=input("tell the passanger is there any other way to go free o
    print("passanger how much we need to pay ")
    chanrges= eval(input("enter the changes by conductor"))
    fare=distance*chanrges
    print(f"fare is {fare}")

elif idcard == 'YES' or 'yes':

    print("No fare will charge")
else:
    print("invalid information")

#####
print("Mom: hey mom")
print("Kid:hello kid")
print("Kid: Mom I got to know that govt. has implemented free bus service")
print("Mom: Yes! I too hear that")
print("Kid: Lets go to Grand mother house")
print("Mom: Okay sure!")
distance = eval(input("Enter a distance: "))
print("Conductor: show me Identity card")
mom_ans = input("Enter your answer: ")
if mom_ans == "no":
    print("Conductor: You need to take the ticket")
    print("Mom: How much fare?")
    print(f"Conductor: you mention the distance {distance} KM for 50Km the
    fare = distance * 100
    print(f"total charge is {fare}")
else:
    print("Conductor enjoy your free trip")

```

```
hey mom
hello kid
kid: Mom I got to know that govt. has implemented free bus services for wo
mens
Mom: Yes! i too hear that
kid: lets go to grand mothers house
Mom: ok, sure
I am conductor of this bus please let me know the distance:- 50
You have the ID card 1. yes or 2. No Show me your Id cardno
tell the passanger is there any other way to go free or have to take the t
icketyes
passanger how much we need to pay
enter the changes by conductor100
fare is 5000
Mom: hey mom
Kid:hello kid
Kid: Mom I got to know that govt. has implemented free bus service
Mom: Yes! I too hear that
Kid: Lets go to Grand mother house
Mom: Okay sure!
Enter a distance: 50
Conductor: show me Identity card
Enter your answer: no
Conductor: You need to take the ticket
Mom: How much fare?
Conductor: you mention the distance 50 KM for 50Km the fare is 100 rs
total charge is 5000
```

In []: